

Pharmacology of General Anesthetics, Part 2

- **Effects and Toxicities of Inhaled anesthetics**
 - **Intravenous anesthetics**
- and adjuncts in anesthesia techniques**

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Dose-dependent effects of inhaled anesthetics on organ systems

	Isoflurane	Desflurane	Sevoflurane	Nitrous Oxide
Cardiovascular (next slide)	↓ blood pressure, little to no effect on cardiac output Depress baroreflex (sevoflurane > isoflurane and desflurane) <i>Desflurane: sympathetic stimulation → tachycardia, HTN</i> Isoflurane / Sevoflurane → ↑HR with increasing doses			Negligible effects (mild ↑ SNS stimulation that offsets myocardial depression)
Respiratory	Respiratory depression (<i>profound with isoflurane</i>) ↓ tidal volume and minute ventilation ↓ ventilatory response to carbon dioxide Bronchodilation (<i>most pronounced with sevoflurane</i>)			Minimal change in minute ventilation and resting arterial pCO ₂
Cerebral	↑ cerebral blood flow and ↑ intracranial pressure* ↓ cerebral metabolic rate <i>Sevoflurane decreases increases CBF and ICP to lesser extent.</i>			↑CBF, ↑ICP ↑ cerebral metabolic rate
Neuromuscular	Relax skeletal muscle Enhance the effects of neuromuscular blockers			Does not relax skeletal muscle or smooth muscle
Smooth muscle	Relax smooth muscle → vasodilation, bronchodilation, uterine smooth muscle relaxation			
Renal	↓ renal blood flow, ↓ glomerular filtration rate, ↓ urine output			The same
Hepatic	↓ hepatic blood flow			The same
Hepatic metabolism	0.2%	<0.1%	5%	0.004%

*↑ Intracranial pressure: Caution in patients with brain tumor, intracranial hemorrhage, or head injury. Anesthetic-related increase in the ICP can cause potentially devastating neurological complications.



Hemodynamic effects of inhaled anesthetics (dose-dependent)

Effects	Comments
Depression of cardiac contractility	Decrease mean arterial pressure in direct proportion to the alveolar concentration of the gas
Vasodilation	Halothane / enflurane more myocardial depression than the others, minimal vasodilation ↓ ↓ CO with little change in SVR Isoflurane, desflurane, sevoflurane ↓ Vasodilation (↓BP) / minimal change in CO ↓ Preserve CO + ↓ afterload, ↓ preload
Takeaway: Isoflurane, desflurane, and sevoflurane have minimal myocardial depressant effects and favorable hemodynamic profiles and may be suitable for patients with impaired myocardial function.	
Desflurane	Transient sympathetic stimulation → tachycardia and hypertension, especially at high doses. Tachycardia may persist. <i>Use with caution and close monitoring in patients with coronary artery disease.</i>

CO: cardiac output; SVR: systemic vascular resistance

➤ At high doses that induce Stage IV anesthesia, **all agents** can cause cardiovascular collapse.



Emergence and Postoperative Phenomena

Effects	Comments
Nausea and vomiting	Drug-induced stimulation of chemoreceptor trigger zone and vomiting center Management: 1) antiemetics: metoclopramide, ondansetron, droperidol 2) avoidance of nitrous oxide (N ₂ O) 3) use of propofol as an induction agent 4) use of ketorolac as a substitute for opioids
Hypertension and tachycardia	↑ Sympathetic tone Sympathetic tone is enhanced by pain.
Post-anesthesia shivering	Result of decrease core temperature (hypothermia) and neurological after-effects of the general anesthetic agent.
Emergence excitement	Tachycardia, restlessness, crying, moaning, and thrashing, and a variety of neurologic signs, including delirium, spasticity, hyperreflexia, and Babinski signs

Toxicities of Inhaled Anesthetics

Toxicities at a glance

	Desflurane	Sevoflurane	Isoflurane	Enflurane	Halothane	Nitrous oxide
Acute toxicity	Low TI	Low TI	Low TI	Low TI	Low TI	Low TI
Malignant hyperthermia	✓	✓	✓	✓	✓	
Hepatotoxicity					✓ “halothane hepatitis”	
Nephrotoxicity		✓ Compound A* vinyl ether cpd		✓ Prolonged exposure		
Carbon monoxide	✓	✓	✓	✓	✓	
Anemia B12 deficiency						✓
Environmental toxicity	✓	✓	✓	✓	✓	✓

Low TI: Low therapeutic Index = Narrow therapeutic window

*No reports of renal injury in humans.



Descriptions of Toxicities

Toxicity	Isoflurane	Desflurane	Sevoflurane	Halothane	Enflurane	Nitrous oxide
Acute	Narrow therapeutic range Acute organ damage is low when used by trained practitioners					
Malignant hyperthermia	Individuals with genetic disorders of skeletal muscle can develop muscle rigidity, hyperthermia, rapid onset of tachycardia and hypercapnia, hyperkalemia, and metabolic acidosis when exposed to succinylcholine or halogenated anesthetic agents					—
Nephrotoxicity	—	—	compound A vinyl ether No renal injury reported	—	Prolonged exposure	—
Hepatotoxicity	➤ Hepatic dysfunction following surgery and general anesthesia is most likely caused by hypovolemic shock, infection conferred by blood transfusion, or other surgical stresses rather than by volatile anesthetic toxicity. Halothane: Fulminant liver failure (1 in 20,000 to 35,000) by direct toxicity and immune response.⁷					

	Isoflurane	Desflurane	Sevoflurane	Halothane	Enflurane	Nitrous oxide
Neurotoxicity	Neural toxicity in infants and young children is an emerging concern. Some, not all, studies suggest increased risk of neurodevelopmental deficits in motor function and behavior. <i>Inadequate evidence to directly attribute the effects to the anesthesia.</i>					
Carbon monoxide production See slide 47	All halogenated anesthetics can produce some carbon monoxide (CO) from their interaction with strong bases in desiccated (dried out) carbon dioxide absorbers . CO binds with hemoglobin with high affinity, reducing oxygen delivery to tissues. Prevention: Fresh carbon dioxide absorbent / avoid dessication <i>Potential for CO formation from most to least: desflurane > isoflurane > > sevoflurane</i>					—
Carcinogenicity	Under normal conditions, inhaled anesthetics are neither mutagens nor carcinogens. No study has demonstrated a causal relationship between anesthetics and cancer.					
Pregnancy	In the US, at least 50,000 pregnant women each year undergo anesthesia and surgery for indications unrelated to pregnancy. The risk of abortion is clearly higher following this experience. It is not obvious whether the underlying disease, surgery, anesthesia, or a combination of these factors is the cause of the increased risk.					
Environment	All inhaled anesthetics are greenhouse gases – hundreds of thousands of times more potent than CO ₂ – They are vented out the buildings into the environment.					

Nitrous oxide's drug-specific effects

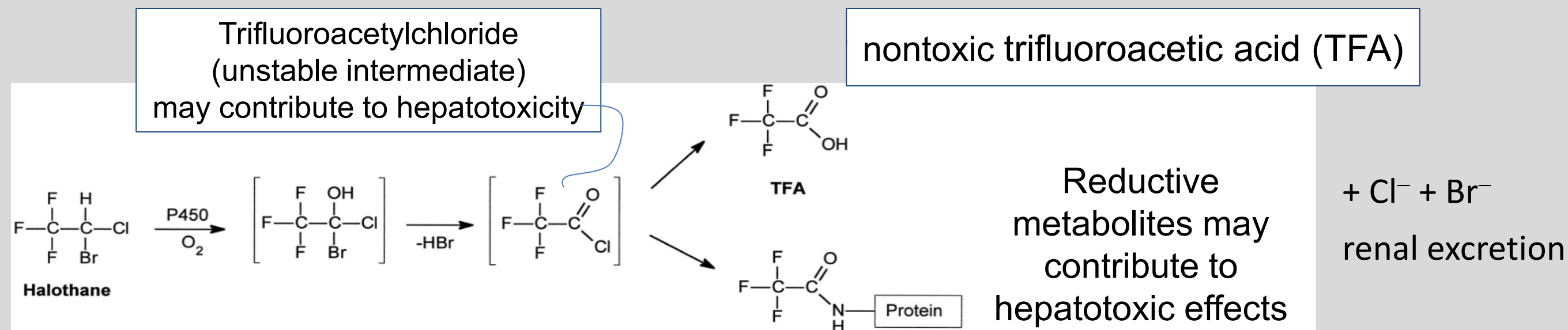
Body space volume expansion	N_2O is ~35x more soluble in blood than N_2 → N_2O diffuses into cavities faster than N_2 diffuses out → increases the volume inside the cavity → expansion of the cavity When walls surrounding the cavity are rigid, the pressure rises instead of the volume.	Cautions: Pneumothorax Intracranial air Middle ear surgery Bowel obstruction N_2O will also diffuse into tracheal tube cuffs, increasing the pressure against the tracheal mucosa.
Vitamin B12 inhibition	Nitrous oxide interferes with vitamin B12 cofactor functions → inactivation of methionine synthase → Signs of vitamin B12 deficiency megaloblastic anemia peripheral neuropathy N_2O should not be used for chronic pain relief (analgesia) or as a sedative in critical care settings. Avoid in patients with disorders of B12, folate, or methionine synthesis or metabolism.	Biochemistry: Vitamin B12 is a cofactor of methionine synthase, which generates methionine from homocysteine. Methionine is necessary for the formation of tetrahydrofolate, which is required for the synthesis of DNA precursors.
Diffusion hypoxia	On discontinuation of N_2O administration, N_2O diffuses from the blood to the alveoli, which dilutes the oxygen in alveoli.	Prevention: 100% O_2 administration instead of air when N_2O is discontinued
Second gas effect The concentration effect of one gas on another	Rapid uptake of N_2O from the alveoli → contraction of alveolar volume associated with the N_2O uptake → concentrates the co-administered inhaled anesthetic remaining in the alveoli → speeds induction	Probably insignificant in the clinical practice of anesthesiology

Halothane Hepatitis (rare): Proposed mechanism

Halothane forms is metabolized by CYP2E1, forming the reactive intermediate trifluoroacetylchloride that binds covalently to liver proteins forming neo-antigens, which induces an immune attack that results in liver cell damage and inflammation.

Halothane hepatitis can rapidly progress to fulminant liver failure.

Risk factors: Genetic predisposition and previous halothane exposure.



Elevated serum aminotransferases can occur with or without prior halothane exposure. Most patients are asymptomatic.

CO₂ absorber in breathing circuit

Carbon dioxide absorbers in closed breathing environments remove carbon dioxide from exhaled breath.

Older CO₂ absorbers in anesthesia breathing circuits consist of strong bases (granular KOH, NaOH) that prevent accumulation of CO₂ in the breathing circuit as the patient exhales.

Desiccated CO₂ absorbers → carbon monoxide (CO) production:

The volatile anesthetics can produce some carbon monoxide from interaction with the strong bases in dry carbon dioxide absorbers.

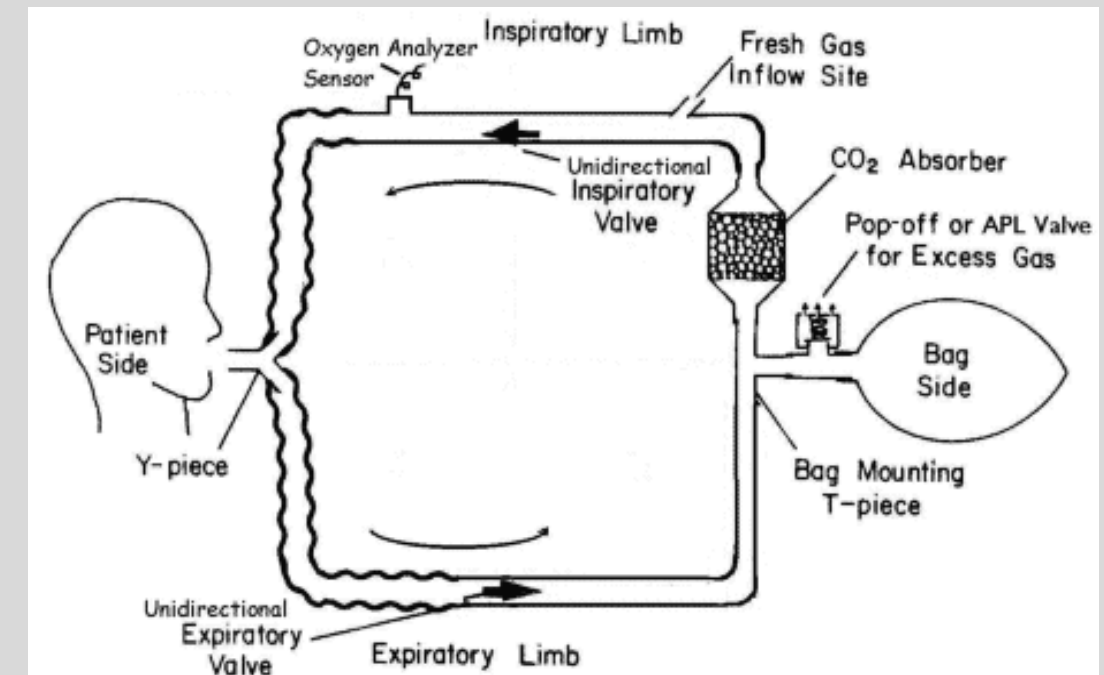
Desiccated absorbers containing stronger alkalis result in greater CO production.

Amount of CO formed at equiMAC doses:

desflurane > enflurane > isoflurane >> sevoflurane

Prevention of CO production:

Fresh CO₂ absorbers should be used and should be changed before they are completely dried out.



Xenon (noble gas)

Xe, atomic number 54

- Colorless, odorless, heavy, inert gas at room temperature and standard pressure
- Obtained by extracting it from the air
- Partition coefficient (λ): $\lambda_{b/g}$ 0.12
- Rapid induction and recovery (extremely insoluble in blood, tissues)
- Noncompetitive antagonism of the NMDA receptor and agonism at the TREK channel (K_{2P} channel, a ligand-gated ion channel present in neurons)
- Minimal side effects
- No cardiovascular effects, no respiratory depression, no known renal or hepatic toxicity
- Not metabolized by human body
- Environmentally safe
- Extremely expensive
- *Not approved in the U.S.*

Check your knowledge

What is the acronym MAC stand for?

How does MAC define potency?

How is MAC used in dosing inhaled anesthetics?

Which inhaled anesthetics are potent volatile agents in common use?

Why is nitrous oxide referred to as a weak anesthetic?

Check your knowledge

Why may isoflurane, desflurane, and sevoflurane be suitable for patients with impaired myocardial function?

What is the concern with the use of desflurane in patients with coronary artery disease?

Which inhaled anesthetics are associated with malignant hyperthermia in patients with a genetic disorder in skeletal muscle calcium regulation?

Check your knowledge

Match the toxicities with the drugs that cause them

Body space volume expansion
Carbon monoxide production in desiccated CO2 absorber
Cortisol synthesis inhibition
Diffusion hypoxia
Environmental toxicity
Malignant hyperthermia
Myoclonus
Nausea and vomiting
Nephrotoxicity (potentially) by compound A
Sympathetic stimulation → tachycardia and hypertension
Vitamin B12 inhibition

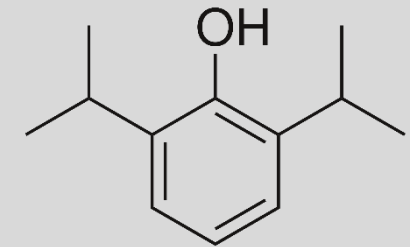
Isoflurane
Desflurane
Sevoflurane
Nitrous oxide

Intravenous Anesthetics

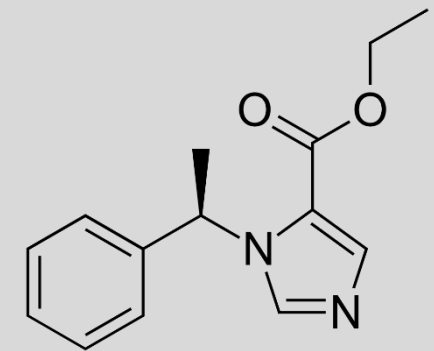
- * Propofol
- * Etomidate
- * Ketamine

Intravenous anesthetics are small hydrophobic, substituted aromatic or heterocyclic compounds.

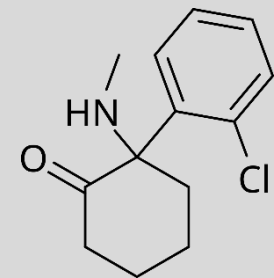
Methohexital is a barbiturate used for induction for electroconvulsive therapy. Availability is limited.



propofol



etomidate



ketamine

Redistribution and intravenous anesthetic's duration of effect

Single bolus dose:

1. Rapid onset:

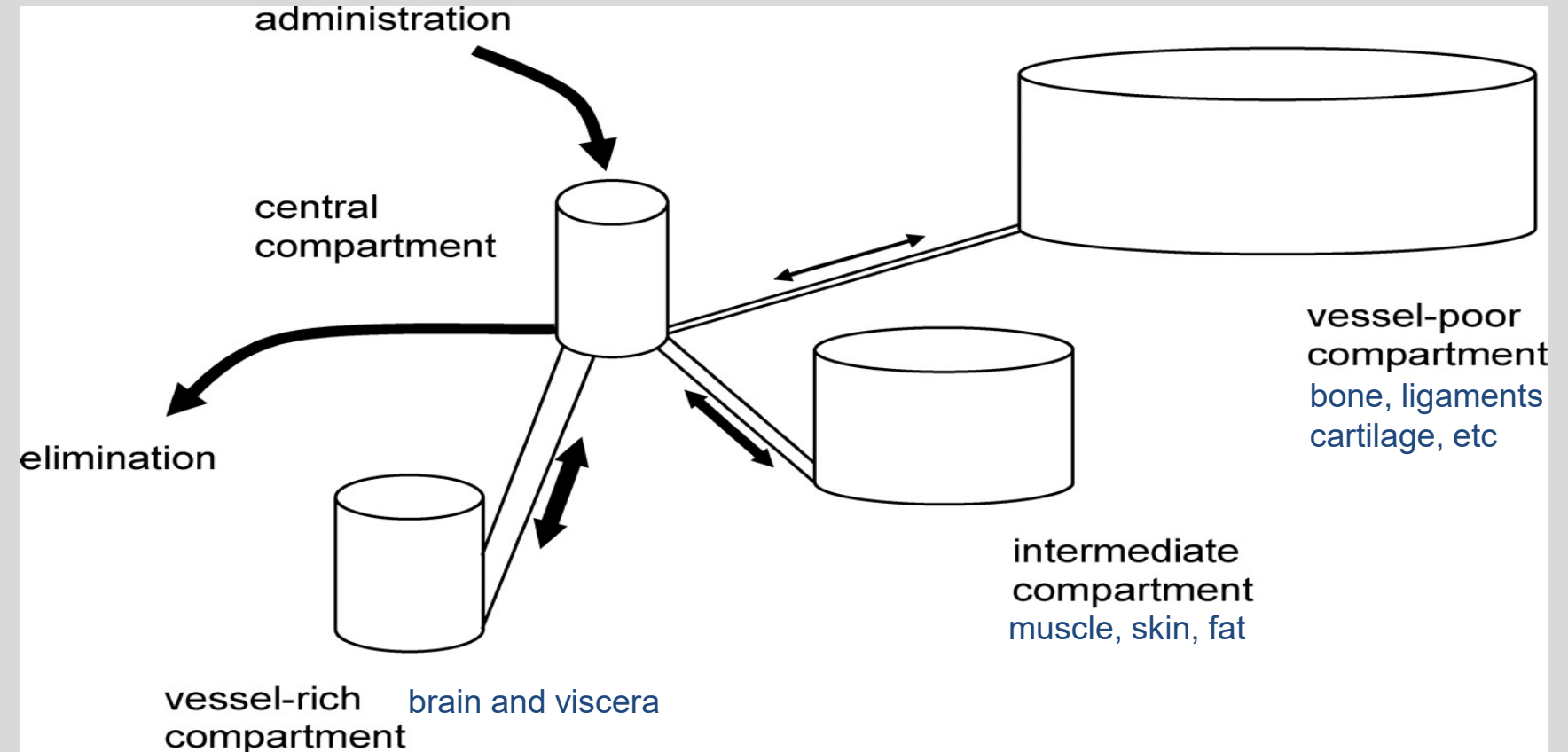
Highly lipophilic IV anesthetics partition into the highly perfused and lipophilic tissues of the brain and spinal cord → anesthesia in a single circulation time.

2. Termination of effect:

As the drug concentration in blood decreases, the anesthetic begins to redistribute out of CNS to less perfused tissues, diminishing the effect → patient emerges from anesthesia

3. Elimination:

Eventually, the drug is metabolized by the liver and excreted by the kidneys.



Three compartment model for a lipid soluble drug.

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Context-sensitive half-time

Half-time: The time required for the concentration of drug to decrease by 50% after stopping a continuous infusion.

Context: The duration of the infusion

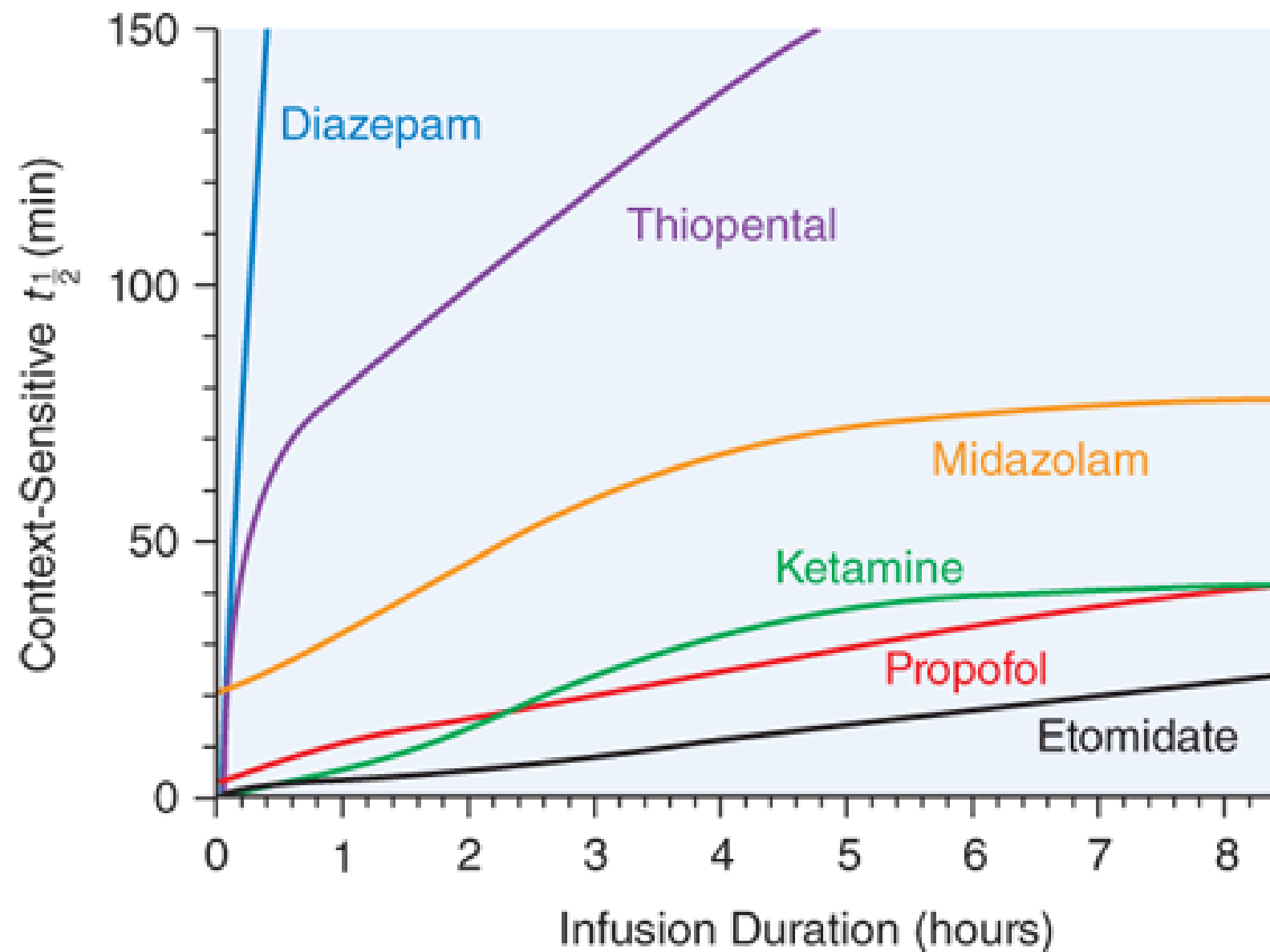
1. IV anesthetic initially distributes to the well perfused tissues.
 2. The drug redistributes to the less perfused tissues over time.
 3. The longer the infusion, the more the drug accumulates in the muscle and fat compartments.
 4. After the infusion is stopped and the drug is metabolized and excreted, blood levels begin to decrease.
 5. The drug redistributes from the peripheral compartments back into the bloodstream.
- For drugs with a long infusion duration, the context-sensitive half-time (time it takes for the concentration of drug in the blood to decrease by one-half) can be significantly longer due to extensive accumulation of drug in peripheral compartments.

Clinical importance:

Context-sensitive half-time helps predict how long it will take the patient to recover after the infusion is stopped.

After prolonged infusions, drug context-sensitive half-times and durations of action are dependent on a complex interaction between the:

- rate of redistribution of the drug,
- amount of drug accumulated in fat,
- and the drug's metabolic rate.
- Propofol, etomidate, and ketamine have brief context-sensitive half-times:
Compared to diazepam and thiopental:
 - they bind less avidly to muscle and fat tissue and
 - they are rapidly metabolized by the liver
- Recovery occurs promptly.
- Propofol is preferred for IV maintenance.
- Adverse effects limit etomidate and ketamine use.
- Diazepam is rarely used in balanced anesthesia. Thiopental manufacture has been discontinued.



Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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I.V. Anesthetics Mechanisms of action and Uses

Drug	MOA	Therapeutic uses
Propofol	GABA-A receptor positive allosteric modulator	• Rapid anesthesia induction • Rapid sequence intubation • Total intravenous anesthesia • Propofol with opioid; other adjuncts for patient-specific or procedure-specific reasons • Sedative-hypnotic in ICU • Propofol • Ketamine, alternative in critically patients without cardiac conditions • Monitored anesthesia care sedation • Propofol
Etomidate	GABA-A receptor positive allosteric modulator	
Ketamine	Noncompetitive NMDA receptor antagonist	



I.V. Anesthetics Mechanisms of action and Uses

Drug	Advantages	Disadvantages
Propofol	Bronchodilation Decreases CMRO ₂ , CBF, ICP Antiemetic / antipruritic Anticonvulsant	Hypotension, dose related Respiratory depression, dose related *Propofol infusion syndrome Painful injection Egg phosphate in soybean oil vehicle (looks like bottled milk): Potential allergens and microbial risk
Etomidate	Hemodynamic stability; no changes in BP, HR, or CO Decreases CMRO ₂ , CBF, ICP Anticonvulsant	High incidence of postop nausea/vomiting Painful injection No analgesic effects Involuntary myoclonic movements Transient acute adrenal suppression Inhibits 11-β-hydroxylase → ↓ cortisol synthesis
Ketamine	Profound analgesic properties Bronchodilation Sympathetic stimulation → maintains CO and MAP without inhibiting respiratory drive (may be beneficial in cardiac instability) Maintains airway reflexes and respiratory drive IM route if IV access is lost	Sympathetic stimulation increases HR, BP, CO and myocardial O ₂ demand (may be harmful in CAD or HTN), Increase pulmonary arterial pressure SBP Potentiates cardiovascular toxicity of cocaine, tricyclic antidepressants Increases CBF, ICP Hallucinations, delirium upon withdrawal, dissociative experiences, nightmares, nausea/vomiting

CMRO₂: cerebral metabolic rate of oxygen; CBF: cerebral blood flow; ICP: intracranial pressure; BP: blood pressure; HR: heart rate; CO: cardiac output; MAP: mean arterial pressure; SBP: systemic blood pressure; CAD: coronary artery disease; HTN: hypertension

*Propofol infusion syndrome, a rare complication: metabolic acidosis, hypotension, hepatomegaly, hyperlipidemia, rhabdomyolysis, CV collapse

Intravenous Agents Used in Monitored Anesthesia Care

*Benzodiazepines | *Propofol | *Ketamine | *Dexmedetomidine | *Opioids | *Lidocaine

Monitored anesthesia care:

A technique without endotracheal intubation in which the anesthesiologist continually monitors and supports the patient's vital functions; diagnoses and treats clinical problems that occur; administers sedative, anxiolytic, or analgesia medications if needed; and converts to general anesthesia if required.

An anesthesia clinician continually monitors and supports the patient's vital functions; diagnoses and treats clinical problems that occur; administers the medications, if needed, and converts to general anesthesia as required.

Drugs with rapid onset and short duration of action are preferred to allow rapid titration to effect and quick recovery.

Benzodiazepines Midazolam	GABA-A receptor positive allosteric modulator · Anxiolysis, sedation, amnesia · Repeated bolus doses or infusions prolongs sedation, confusion, and amnesia into the postop period · Respiratory depression · Reversal agent: Flumazenil
Propofol	GABA-A receptor positive allosteric modulator · Amnestic and antiemetic properties at subhypnotic doses · Depth of sedation can be rapidly adjusted. · Narrow therapeutic window; patients may rapidly transition to deeper sedation. · Respiratory depression · No reversal agent

Ketamine	Noncompetitive NMDA receptor inhibitor that blocks glutamate • Dissociative state, amnesia, profound analgesia that can decrease need for opioids • Minimal respiratory depression • Central sympathetic stimulation; ↑HR, BP, cardiac output • Intrinsic myocardial depression; cardiac depressant in catecholamine-depleted patients • Salivation that can cause laryngospasm • Emergence phenomena: hallucinations, vivid dreams, out-of-body experiences
Dexmedetomidine	Highly selective the α_2-adrenergic receptor agonist which inhibits sympathetic actions at brainstem and spinal cord • Sedative, anxiolytic, analgesic (modest) properties • Does not provide reliable amnesia or anesthesia • Slower onset of sedation (10-15 min) • Less respiratory depressant effect than other sedatives • Bradycardia, hypotension, dry mouth

Opioids: Fentanyl, Remifentanyl (short-acting)	Mu opioid receptor agonists • Analgesia: increase pain threshold, alter pain perception, and inhibit ascending pain pathway • Cause intraoperative and postoperative nausea and vomiting • Respiratory depression • Reversal agent: Naloxone
Lidocaine IV infusion	Voltage-gated sodium channel blocker • May be used during induction of general anesthesia to suppress the cough reflex and laryngospasm, reduce the pain of injection or other induction agents, and supplement anesthetic effects. • Lidocaine-induced ventricular rate can contribute to arrhythmia. Arrhythmias are uncommon when IV lidocaine is used for pain control. • Lidocaine infusion requires close monitoring for signs of cardiac and local anesthetic systemic toxicity(LAST)

Summary

Anesthetic	B:G	B:B	¹ MAC	Comments
Nitrous oxide	0.47	1.1	>100	Weak anesthetic; rapid onset and recovery Not metabolized Can cause expansion of air-fill cavities / Diffusional hypoxia Prolonged exposure can result in megaloblastic anemia, peripheral neuropathies
Desflurane	0.42	1.3	6-7	Low volatility; rapid onset/recovery; poor induction agent (respiratory irritation) SNS stimulation → transient, often marked, tachycardia, ↑BP Carbon monoxide can be produced from desiccated CO ₂ absorbents in breathing circuit
Sevoflurane	0.69	1.7	2.0	Rapid onset/ recovery Smooth rapid inhalation induction; bronchodilating – preferred for patients with airway problems Hepatic metabolism (~5% of absorbed drug) Interaction with soda-lime → Compound A vinyl ether (renal toxicity not reported)
Isoflurane	1.40	2.6	1.4	Medium rate of onset and recovery; pungent odor → poor induction agent; dilates coronary arteries
Enflurane	1.80	1.4	1.7	Medium rate of onset / recovery; limited biodegradation; seizure potential
Halothane	2.30	2.9	0.7	Relatively slow onset / recovery; significant hepatic metabolism; halothane hepatitis – reactive intermediary – direct toxicity, immune reaction

¹MAC=median alveolar concentration in vol%: the anesthetic concentration that produces immobility in 50% of patients exposed to noxious stimulus; MAC is small for potent anesthetics and large for less potent anesthetics.

B:G=blood:gas partitions coefficients; B:B=brain:blood partition coefficients

Adapted from Basic & Clinical Pharmacology Table 25-2



Common advantages and disadvantages of potent volatile anesthetics

Advantages

- Ease of administration
- Reliable block of sensory input (ie, pain)
- Bronchodilation
- Decrease in skeletal and smooth muscle tone (dose-dependent)
- Decreased cerebral metabolic rate
- Increased cerebral blood flow by direct vasodilatory effect
- Monitoring end-tidal anesthesia concentration as a measure of anesthetic depth

Disadvantage

- Dose-dependent...
 - Suppression of airway reflexes
 - Respiratory depression
 - Hypotension due to myocardial depression and vasodilation
- Increased post-op nausea/vomiting compared to IV anesthetics (antiemetic needed)
- Malignant hyperthermia potential in susceptible individuals

Nitrous Oxide: Summary of effects

Anesthetic effects	Incomplete (weak) anesthesia ➤ Adjunctive use with inhalation and intravenous anesthetics for sedation, analgesia, and anxiolytic effects
Muscle effects	Does not relax skeletal muscle → does not enhance the effects of neuromuscular blocking drugs Does not relax uterine smooth muscle ➤ Does not cause malignant hyperthermia
Organ effects	Sympathetic stimulation → maintains hemodynamics Increases respiratory rate Reduces hepatic and renal blood flow not hepato- or nephrotoxic
Other effects	• Body space volume expansion and increased pressure in spaces with noncompliant walls. • Interferes with vitamin B12 function; avoid in patients with disorders of B12 deficiency and folate or methionine synthesis • Diffusion hypoxia • Second gas effect (concentration effect)

IV Agents	MOA	Benefits	Adverse effects	Notes
Propofol	GABA-A positive allosteric modulator	Excellent sedation Bronchodilation Antiemetic Antiseizure Antipruritic	Dose related hypotension No absolute contraindications	Ultra-short-acting Frequent use for maintenance of anesthesia during total IV anesthesia
Etomidate	GABA-A positive allosteric modulator	Excellent sedation with little hypotension	Suppresses adrenal function Myoclonus (not related to seizure)	Ultra-short-acting Caution in sepsis Initial dose of glucocorticoid may be needed
Ketamine	Blocks excitation by glutamate at NMDA receptors	Stimulates catecholamine release Bronchodilation excellent Maintains airway reflexes and respiratory drive Analgesia excellent	Sympathetic stimulation→↑HR, BP, cardiac output	May be preferred in patients with bronchospasm, septic shock and hemodynamic compromise Caution in ischemic heart disease



IV Agents	MOA	Benefits	Adverse effects	Notes
Midazolam	GABA-A positive allosteric modulator	Sedation Anxiolytic Anterograde amnesia Antiseizure	Mild vasodilation, ↓BP and cardiac output Respiratory depression with concurrent opioid	Not frequently used for deep sedation – propofol and etomidate have superior effectiveness
Opioids	Mu opioid receptor agonists	Analgesia Supplemental sedation Suppress airway reflexes during intubation Reduce stress responses	Respiratory depression Bradycardia Exacerbate hypotension by sedative-hypnotic Nausea, vomiting, ileus	Common use Short-acting opioids during induction
Lidocaine IV infusion	Blocks voltage-gated sodium channels in nerve cells	Suppress the cough reflex laryngospasm Reduces pain of injection or other induction agents, and supplement anesthetic effects Decreases need for opioids	Lidocaine-induced ventricular rate can contribute to arrhythmia. Arrhythmias are uncommon when IV lidocaine is used for pain control.	Lidocaine infusion requires close monitoring for signs of cardiac and local anesthetic systemic toxicity(LAST)

Check your knowledge

What is the fundamental concept that ensures that the anesthetic concentration in the CNS is stable and effective for maintaining anesthesia during the procedure?

- When the partial pressure of the anesthetic gas in the inspired air, alveoli, arterial blood, and CNS are in equilibrium (steady state, $F_A/F_I=1$), the desired clinical effect will be produced.

What four factors influence the uptake of anesthesia?

- Inspired concentration of anesthetic gas
- Solubility of gas in blood (defined by the blood:gas partition coefficient)
- Cardiac output
- Alveolar-venous partial pressure difference

Why does faster the uptake of anesthetic agent lead to a slower speed of induction?

- When uptake is fast, a large amount of anesthetic is absorbed into the blood from the alveoli → creates a significant difference between the inspired concentration and the alveolar concentration ($F_A < F_I$) as the alveoli are depleted of anesthetic faster than they can be replenished by inhalation → it takes longer to reach equilibrium ($F_A/F_I=1$)

Check your knowledge

Why is the speed of induction slower with the more soluble anesthetic gases?

- The more soluble the anesthetic is in blood (the higher the partition coefficient) → the greater the capacity for uptake into the blood → the slower rate of rise in partial pressure → the longer it takes to reach equilibrium

How does an increase in cardiac output lead to a slower speed of induction?

- ↑ cardiac output → ↑ pulmonary blood flow → the partial pressure of the gas in blood rises more slowly → slower rise in F_A/F_I towards equilibrium → slower speed of onset
- This effect is more pronounced with soluble agents, which achieve equilibrium slower.

How does a large alveolar to venous partial pressure difference influence the speed of induction?

- Anesthetic gas is taken up from arterial blood into nonneural tissues → venous circulation returns to pulmonary circulation with little or no dissolved anesthetic gas → generates a partial pressure difference between alveolar gas and pulmonary blood.
- The greater the difference in gas concentrations, the longer the time to achieve equilibrium with brain tissue.

Check your knowledge

What is the acronym MAC stand for?

- Minimum alveolar concentration

How does MAC define potency?

- 1 MAC = the concentration of the inhaled gas in the alveoli that prevents movement in 50% of patients in response to a standardized stimulus (eg, surgical incision) at standard temperature and ambient pressure.
- MAC reflects brain partial pressure and allows comparisons of potency between agents.

How is MAC used in dosing inhaled anesthetics?

- Dosing of inhaled anesthetics is guided by MAC and MAC values are additive.

Which inhaled anesthetics are potent volatile agents in common use?

- Isoflurane, desflurane, sevoflurane

Why is nitrous oxide referred to as a weak anesthetic?

- Nitrous oxide has low potency. Even 100% nitrous oxide does not achieve 1 MAC. Therefore, it is combined with a potent volatile anesthetic.

Check your knowledge

Why may isoflurane, desflurane, and sevoflurane be suitable for patients with impaired myocardial function?

- Isoflurane, desflurane, and sevoflurane cause vasodilation, which lowers systemic vascular resistance and blood pressure with minimal change in cardiac output (preserves cardiac output) – of course, with close monitoring to ensure patient safety and optimal outcomes.

What is the concern with the use of desflurane in patients with coronary artery disease?

- Although desflurane causes vasodilation with minimal effect on the force of contraction (inotropy), its transient sympathomimetic properties cause tachycardia and hypertension, especially at high dose. These cardiac effect may induce cardiac ischemia and heart attack.

Which inhaled anesthetics are associated with malignant hyperthermia in patients with a genetic disorder in skeletal muscle calcium regulation?

- The halogenated anesthetics: isoflurane, desflurane, sevoflurane, halothane, enflurane
- Not nitrous oxide.

Check your knowledge

Match the toxicities with the drugs that cause them

Body space volume expansion	Nitrous oxide
Carbon monoxide production in desiccated CO ₂ absorber	Halogenated volatile anesthetics
Cortisol synthesis inhibition	Isoflurane
Diffusion hypoxia	Nitrous oxide
Environmental toxicity	All inhaled anesthetics
Malignant hyperthermia	Halogenated volatile anesthetics
Myoclonus	Isoflurane
Nausea and vomiting	All inhaled anesthetics
Nephrotoxicity (potentially) by compound A	Sevoflurane
Sympathetic stimulation → tachycardia and hypertension	Desflurane
Vitamin B12 inhibition	Nitrous oxide

References

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- Access Medicine Katzung's Basic & Clinical Pharmacology 16e, 2024; Chapter 25: General Anesthetics
- Access Medicine Morgan & Mikhail's Clinical Anesthesiology, 7e, 2022; Chapter 7: Pharmacological Principles; Chapter 8: Inhalation anesthetics; Chapter 9: Intravenous Anesthetics; Chapter 52: Thermoregulation, Hypothermia, & Malignant Hyperthermia
- Lexidrug monographs
- UpToDate various articles; literature current through Dec 2024

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